

Leveraging Directed Acyclic Graphs for Advanced Ensemble Learning in Predictive Maintenance for Industry 4.0

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Abstract—In Industry 4.0, Predictive maintenance is important because it helps to minimize operating time wastages due to equipment failures. This paper proposes a novel approach using Directed Acyclic Graphs (DAGs) to ensemble multiple machine learning models, enabling more robust and accurate failure predictions. We demonstrate how DAGs can be used to dynamically structure the flow of forecasts, with models feeding into each other to form a sophisticated prediction pipeline. This approach allows multiple complex model interactions, improving maintenance scheduling and anomaly detection in industrial settings. A comparative analysis of DAG-based ensemble learning is presented against traditional methods such as bagging and boosting.

Keywords: Directed Acyclic Graphs, Ensemble Learning, Predictive Maintenance, Industry 4.0, Machine Learning, AI, IoT

I. INTRODUCTION

Industry 4.0 predictive maintenance utilizes machine learning techniques to mine sensor data to predict equipment failures before they lead to more extensive downtimes. They include first-order methods such as Random Forest (RF), gradient boosting machines (GBM), and extreme gradient boosting (XGB). However, their static architectures may only embody part of the relation model of complex industrial equipment systems. Much of the work has been based on traditional forms of ensemble learning known as bagging, boosting and stacking. While useful for cascading models, the ideas lack dynamic information flow between classifiers. Given this, more sophisticated methods that can accommodate inter-classifier dependencies while modeling would be more effective in approximating complex real-world systems. These fixed structures can be suboptimal while working with complex data since one or another model may work better at different features of a given data set or may produce better results when operated in specific modes. The problem of not knowing how to design more flexible structures for ensembles to address such complexities is another significant research question. Also, there is a scarcity of research explaining which new ensembling approaches, besides essential Voting and Stacking, can be used in the context of predictive maintenance in Industry 4.0 and how the new approaches work in terms of processing

the data through layers of models connected by Directed Acyclic Graphs.

This research aims to introduce one such ensemble approach to machine learning models known as directed acyclic graph (DAG). While cascade models specify linear relationships between classifiers, DAGs can have any number of input classifiers for a single classifier, thus allowing a model to input others. Following this structure, the authors intended to boost the effectiveness of maintenance approaches and increase the reliability of discovering equipment failures at industrial organizations to minimize unexpected blackouts. Potential contributions include using DAGs to enhance model aggregation methods and maintenance scheduling to reduce operating disruptions. One identified innovation is using DAG-based ensembling for predictive maintenance problem areas, which needs further investigation. Based on the analysis above, this DAG structure leverages flexibility compared to ordinary ensemble models and is suitable for complex industrial data. Therefore, we show how using the variance of DAG-based ensembles and standard methods allows for a better prognosis. This introduces a new DAG-based structure for modeling ensembling for Industry 4.0, which can be adapted to other industrial deep-learning AI issues that rely on complex dynamic structures.

II. LITERATURE REVIEW

A cursory look at the current literature on predictive maintenance methodologies shows that Industry 4.0 heavily relies on ensemble learning methods for analyzing large volumes of sensory data for detecting equipment faults. Essentially, ensemble methods integrate multiple machine learning models to create a more robust predictive model than any of the models that make up the ensemble. Of all the ensemble learning methods, the simplest tree-based aggregation of bags, boosting, and stacking have been widely researched for equipment failure predictions employing the high number of signals from sensors in Industry 4.0 environments.

Bagging (bootstrap aggregating) is a technique that involves training a number of base models, mostly decision tree models, on different bootstrap samples. This technique helps decrease variation from the base models, which have

been trained separately, by arriving at the average of the models involved. For the hard threshold, only new models with better prediction of the current ensemble models on poorly predicted instances are added to the ensemble. This puts model capacity on a trajectory towards performing well in the problematic areas of the feature space. Stacking employs a second meta-model to train how individual predictions derived from the primary base learners in the ensemble are combined.

Specifically, random forests and the recently developed gradient-boosted decision trees of trees-based methods have been used for failure prediction in Industry 4.0 applications. They are amenable data models for the semantically diverse sensor data streams produced and also permit easy explanation of forecasts for use in decision-making for predictive maintenance. The availability of different data types and the necessity to explain the predictions distinguish tree ensembles for the considered applications. More work on how different ensemble techniques can be adapted to suit the type of data and the kind of operational hindrances discovered in Industry 4.0 sensor-enabled intelligent factories can be helpful for extra improvement. Other improvements in computation speed, addressing drift and online learning are desirable avenues that could make the ensemble predictive maintenance systems even more appropriate for the extensive real-time data flows and dynamic nature of manufacturing environments. For example, Breiman's Random Forest algorithm [1] has been shown to improve classification accuracy by averaging the predictions of multiple decision trees, reducing the variance of individual classifiers. Also, the Gradient Boosting algorithm developed by Friedman [2] is used to improve the prediction that sequentially corrects the mistakes of weak learners, and it has been successfully implemented for industrial predictive maintenance [3]. However, these methods, mainly based on the ensemble strategy, provide a relatively strict and predetermined structure and thus are not so suitable for solving the highly coupled and complicated problems of newly emerging industrial systems to process highly correlated and complex data. In the context of Industry 4.0, PM has witnessed profound enhancement because of coupling machine learning models with real-time IIoT data streams. Kong et al. [4] implemented different ensemble machine learning techniques for condition monitoring and prediction of the remaining useful life of wind turbines, wherein using SVM and Random Forest as an integrated framework confirmed that ensemble learning techniques can be efficiently applied to deal with the time series data. However, this study adopted simple stacking techniques in which the models were layered rigidly, which reduced the possibility of data exchange between classifiers. For instance, Sun et al. [5] applied ensemble methods to predictive maintenance for industrial robots presented with a decision tree and neural network. Beyond conventional ensemble methods, recent research has begun exploring more advanced model ensembling techniques. Krawczyk et al. [6] examined ensemble learning for imbalanced data, a common issue in predictive maintenance where failure events are rare.

Similarly, Botezatu et al. [7] presented a deep learning ensemble for predictive maintenance in hard drive failures, achieving impressive results but using fixed ensembling strategies again. DAGs have also been popular in machine learning, where they were extensively reviewed in the design of neural networks and, more so, in causal inference. Pearl's work [8] with causal graphs and DAGs directly set the underpinnings for how dependencies between variables can have the structure in a noncyclic, directed form. Although this work mainly deals with causal modeling, it has sparked the use of DAGs in the architecture of intricate machine learning processes. More recently, Ghosh et al. [9] used DAG-based architectures in deep learning pipelines in healthcare applications and demonstrated how such models make the models more understandable and provide flexibility in information flow between models.

Socially, Saeed et al. [10] employed DAG-based neural networks for anomaly detection in IoT networks to illustrate how complex model structures enhance DAG networks since the flexible structures can incorporate interdependencies in streaming data related to predictive maintenance. In predictive maintenance specifically, Zhou et al. [11] reviewed dynamic ensemble methods, which adapt the model combination based on changing data distributions.

Similarly, Zhang et al. [12] explored adaptive boosting algorithms in predictive maintenance, where the ensemble adapts as new sensor data.

Finally, in predictive maintenance for Industry 4.0, models must handle the prediction task and the streaming nature of IoT data. A recent study by Li et al. [13] proposed an IoT-based predictive maintenance system using a hybrid ensemble of Random Forest and Convolutional Neural Networks (CNNs). On this basis, Zhao et al. [14] proposed an adaptive ensemble model for predictive maintenance based on the application of neural networks and decision trees. Still, they did not address DAG-based architectures, which could give more exciting interactions between models.

Although using ensemble methods has been widely explored in predictive maintenance, most methodologies are based on fixed architectures, such as stacking or boosting; hence more flexible than needed to capture all peculiarities of IIoT data. This problem is solved through Directed Acyclic Graphs (DAGs), which provide a better mechanism to handle data flow between models, preventing the establishment of a rigid chain-like structure that cannot accommodate complex dependencies and relationships associated with the data. This seems to be where this work can fill in the gap by proposing an ensemble learning framework based on a DAG that is well suited for predictive maintenance in Industry 4.0 compared to other generic ensemble approaches.

III. METHODOLOGY

A. Problem Definition

Relative to the Industry 4.0 concept, predictive maintenance refers to determining possible equipment breakdowns before they happen so that maintenance

procedures can be scheduled. This is especially so in industries where machinery is critical to operations, and failure can lead to loss-making. Predictive maintenance tasks are usually presented as classification or regression problems in which the machine learning models will predict whether a machine will fail within a given time horizon into the future or estimate the RUL.

Let $D = \{(X_1, y_1), (X_2, y_2), \dots, (X_n, y_n)\}$, where X_i represents the feature vector, for instance, i (sensor data such as temperature, pressure, vibration) and $y_i \in \{0, 1\}$ represents the class label indicating whether a failure will occur, the goal is to learn a function $f: X \rightarrow y$ that offers a precise estimate to equipment failures. Conventional approaches like the classic machine learning methods, including the Support Vector Machines (SVMs) or Random forests, are efficient but are inherently rigid structures that may not be able to capture the intricate dependency in [IoT data] from the industrial domain.

To overcome these challenges, the present paper aims to address an innovative solution: an ensemble method based on a Directed Acyclic Graph (DAG). This approach offers significant advantages in terms of a well-proportioned structure of putting together several models and using all advantageous aspects of each. This increases the flexibility of the flow of information between models, which can provide results superior to traditional linear methods on complex industrial datasets.

B. Detailed Architecture of DAG Ensemble

The concept behind the Directed Acyclic Graph (DAG) Ensemble architecture is that one strong base AI model is selected from a group of such models. In this architecture, the nodes are essentially AI models, and a directed edge indicates that information flows from one model to another. The acyclic structure is the main focus of this architecture since it does not allow information flow loops. This section overviews the DAG Ensemble architecture and uses the example of the predictive maintenance application.

C. Base Model Selection and Placement

The DAG Ensemble is built with a range of unrelated base AI models, including RF and SVM, NN and XGBoost, among others. These models are selected by focusing on these merits to identify how they can extract different patterns from the data. Random Forest has high interpretability and low sensitivity to overfitting, but Neural Networks can generate non-linear models. The placement of these models is structured in a layered fashion:

- **Input Node:** The first node in the architecture is the input node that takes raw data fed into the ensemble, including temperature, pressure and vibration data.
- **Base Model Layer:** The input node passes the data to a set of base models—Random Forest, SVM, NN, and XGBoost—through directed edges. Each of these models processes the input independently, producing individual predictions.
- **Intermediate Node (Optional):** In some cases, the outputs of specific base models are aggregated through an intermediate node before reaching the final decision. For instance, the Random Forest

and SVM outputs might be combined using a meta-learner, such as a Logistic Regression Model, to generate an intermediate prediction.

- **Final Decision Node:** The final Node aggregates the outputs of all base models (or the intermediate nodes) to produce a single prediction. This aggregation can be performed using majority voting or weighted averaging techniques.

D. Concrete Example: Predictive Maintenance in Turbines

To illustrate the DAG Ensemble architecture, the predictive maintenance scenario for turbines can be considered. Based on sensor data, the goal is to predict whether a turbine will fail.

For this task, a DAG topography, as shown in Figure 1, could be developed where:

- The input data, consisting of sensor readings (e.g., temperature and pressure), is first passed to the node for feature selection. This node could be implemented using PCA, AutoEncoder or a similar model. This node generates lower-dimension data fed to the base model nodes and implemented by models like Random Forest, SVM, NN, and XGBoost through directed edges.
- Each model (individually trained by a separate training module) processes the incoming data and makes a prediction. Suppose the projections are as follows: Random Forest predicts “no failure,” SVM predicts “failure,” Neural Networks predicts “failure,” and XGBoost predicts “no failure.”
- These predictions are passed to the optional intermediate node, where meta-learning occurs through models like logic regression, MAML etc. The base model outputs are clustered for meta-learning following a specific strategy that depends on the task. In the figure, Random Forest & SVM outputs have been grouped to be meta-learned through logic regression and the rest are clustered to be meta-learned through MAML. This meta-learning layer is entirely optional.
- The final decision node, the Final Ensemble, aggregates the predictions of meta-learning nodes. The final ensemble employs any selected ensembling strategy, such as Weighted Average or Max Voting, and outputs the ensemble’s prediction to the outside world.

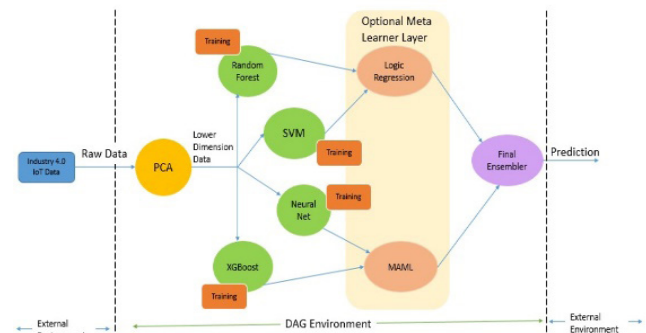


Fig. 1: Schematic Diagram of DAG Ensemble for Predictive Maintenance

E. Significance of Edges in the DAG

The DAG ensemble architecture has machine-learning models nodes, and the edges indicate the flow of the information from one node to another. In particular, raw input data flows only in the forward direction from the input node to several HBLs, such as random forest, support vector machine, multilayered neural network, gradient-boosted decision tree HBL, etc. The above base models then transmit their estimations to intermediate nodes across directed arcs, and it is now easily aggregated without using circular reasoning. The DAG connectivity thus allows numerous models to take the local decisions and pipeline them into a set of unified outcomes in parallel using the benefits of ensemble diversification without aggravating error cumulative chain results. In a broader sense, graphical ensemble architectures enable the tunable model architectures that employ the two gross types of ensemble integration, vertical and horizontal, to take full advantage of heterogeneity with learners and inputs.

F. Model Aggregation and Final Decision

The aggregation of model outputs can be done through several strategies. In the example above, majority voting was used at the intermediate Node to combine the predictions of the base models. Alternatively, the aggregation could be weighted, where models with better historical performance contribute more significantly to the final decision. In cases where no intermediate node is used, the base models' predictions are directly aggregated at the final decision node. This makes projections layered and structured more accessible for interpretation. The possibility of overfitting is also eliminated since the structure is directed acyclic. Due to the ability to use many models at once, the DAG Ensemble is a very effective tool for increasing the accuracy of predictions, even in such cases as predictive maintenance.

G. Dataset

The proposed DAG-based ensemble is evaluated using two well-known datasets for predictive maintenance: The two datasets used in this paper are the NASA Turbofan Engine Degradation dataset [15] and the PHM Society's 2008 Data Challenge dataset [16].

- **NASA Turbofan Engine Degradation Dataset:** This dataset consists of sensor data from several turbofan engines for their Remaining Useful Life (RUL) prognosis. The given dataset also documents the pressure, temperature, and vibrational measurements of various operating conditions. It is suitable for our analysis because the performance deteriorates gradually, emulating authentic industrial IoT data.
- **PHM Society's 2008 Data Challenge Dataset:** This dataset contains sensor attributes for industrial systems emphasizing fault identification and failure prognosis. It offers a diverse environment for developing and evaluating supervised learning algorithms for predictive maintenance problems and offers multiple types of sensors and failures.

Given all the information and by evaluating the DAG-

based ensemble on these datasets, we can show how this model can help predict malfunctions in an assembly line of industrial machines and operation conditions.

H. Ensemble Strategy

The ensemble employs a combination of meta-learning and voting strategies to aggregate model outputs:

- **Meta-Learning:** The meta-classifier (XGBoost or logistic regression) acquires the optimal manner in which the outcomes of the base classifiers are to be combined. This method enables the base models with higher prediction accuracy to dominate the results and enhance the final result of the ensemble.
- **Voting:** It is in situations where several classifiers come up with different or simultaneous forecasts, as in Random Forest, SVM, Neural Network, etc., where the estimates are evaluated employing a simple majority rule or weighted vote. This allows the ensemble to take advantage of the strengths of different models without necessarily fixing the combination ratio.
- **Stacking:** The responses generated from the base models are then forwarded to a higher-order learner who makes the final decision. This strategy allows the ensemble to mimic the interaction of different model outputs more effectively.

I. Performance Metrics

To assess the performance of the DAG-based ensemble, several metrics are used:

- **Accuracy:** The ratio of the actual positive values predicted correctly by the model.
- **Precision and Recall:** Precision checks the quantity of genuinely positive results that are assigned out of all the positive results that the model, on the other hand, predicted; recall measures the proportion that is genuinely positive out of the overall positive results.
- **F1-Score:** The average stratified mean of precision over all the classes or the average stratified mean of recall over all the classes, offering a single number that can reflect the balance between the two sides of the model's classification capabilities.
- **ROC-AUC:** The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) summarizes the model's ability to distinguish between positive and negative classes.

These metrics comprehensively evaluate the proposed ensemble's performance on the NASA and PHM datasets, allowing comparisons with other predictive maintenance models.

IV. EXPERIMENT

A. Experiment Objective

More specifically, the primary objective of this experiment is to provide a detailed investigation of a newly introduced predictive approach in fault prognosis in manufacturing systems in the Industry 4.0 environment. The proposed DAG maintenance modeling and fault prediction performance will be tested in an empirical study to other

standard maintenance modeling methods like random forest, gradient boosting machines, support vector machines, and artificial neural networks trained separately. Specifically, we will compare the proposed DAEG-FPD model to an existing method of an ensemble that employs the stacking or blending of the base learner models to enhance the failure prediction capability. Specifically, we aim to assess how well this model performs in predicting equipment failures compared to traditional ensemble methods, such as Random Forests and Gradient Boosting, as well as standalone models like Support Vector Machines and Neural Networks. By analyzing various performance metrics including accuracy, precision, recall, F1-score, and ROC-AUC—we seek to demonstrate the benefits of a flexible, multi-model approach to predictive maintenance.

B. Experiment Setup

The experiment was conducted on a 64-bit Linux system containing a 2-GHz processor and 13 Gb RAM to realize the DAG Ensemble. More specifically, experiments employed the assistance of Python language with the sci-kit-learn, Keras, XGBoost, and NumPy toolkits. Two datasets were used: the NASA Turbofan and the PHM Society 2008 datasets. A 70:30 between-train-test split ensured the models were tested with data they hadn’t seen before. The DAG Ensemble was compared to single classical models – Random Forest, Support Vector Machines (SVM), Neural Networks, and XGBoost.

C. Experiment Results

The results of the experiment are presented in Table I. This table summarizes the performance metrics for the DAG-based ensemble and the baseline models.

Table I: Performance Metrics of Models

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
Random Forest	0.85	0.80	0.82	0.81	0.90
SVM	0.82	0.78	0.76	0.77	0.88
Neural Network	0.83	0.79	0.80	0.79	0.89
XGBoost	0.87	0.83	0.85	0.84	0.91
DAG Ensemble	0.90	0.88	0.87	0.87	0.94

Additionally, Figure 2 visually represents the performance metrics, showing all the parameters for each model.

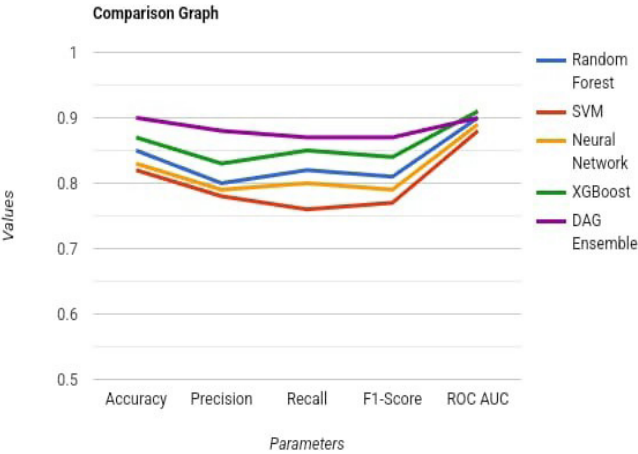


Fig. 2: Comparison of Performance Metrics Among Models

To ensure an accurate evaluation, confusion matrices were generated for the DAG ensemble model and the baseline models using the entire test set from the NASA Turbofan Engine Degradation dataset, which contains 5163 data points. The confusion matrices for each model are shown in Tables II, III, IV, V, and VI.

Table II: Confusion Matrix for DAG Ensemble Model

	Predicted: 0	Predicted: 1
Actual: 0	2500	180
Actual: 1	45	2438

Table III: Confusion Matrix for Random Forest Model

	Predicted: 0	Predicted: 1
Actual: 0	2400	280
Actual: 1	100	2383

Table IV: Confusion Matrix for SVM Model

	Predicted: 0	Predicted: 1
Actual: 0	2300	380
Actual: 1	190	2293

Table V: Confusion Matrix for Neural Network Model

	Predicted: 0	Predicted: 1
Actual: 0	2350	330
Actual: 1	140	2343

Table VI: Confusion Matrix for XGBoost Model

	Predicted: 0	Predicted: 1
Actual: 0	2450	230
Actual: 1	90	2393

Table VII and heatmap Figure 3 summarize each model’s confusion matrix results to facilitate comparison across the different models. The comparison is also expressed through a bar graph in Figure 4.

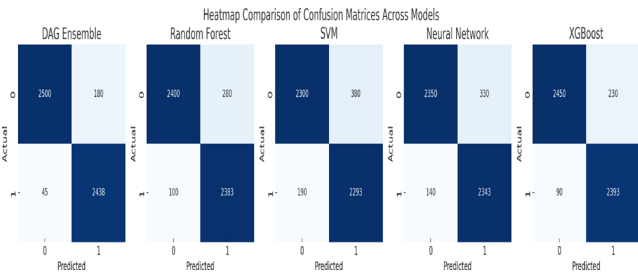


Fig. 3: Heatmap Comparison of Confusion Matrices Across Models

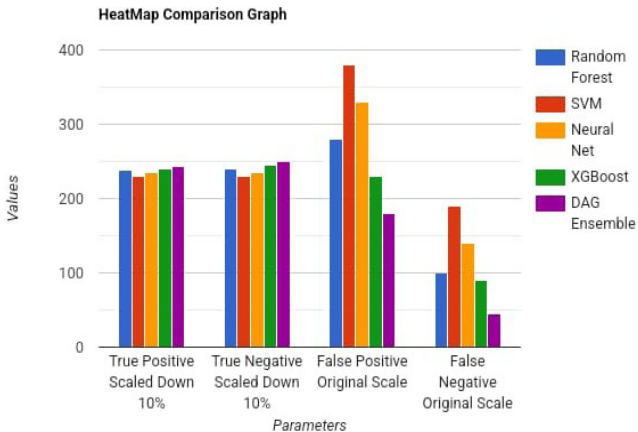


Fig. 4: Bargraph Comparison of Confusion Matrices Across Models

The confusion matrix analysis demonstrates, as shown in Figure 4 that the DAG ensemble model offers a more reliable solution for predictive maintenance. It minimizes false alarms and missed failure detection, crucial in industrial applications where downtime or failures can be costly.

Table VII: Summary of Confusion Matrix Results

Model	True Positives (TP)	True Negatives (TN)	False Positives (FP)	False Negatives (FN)
Random Forest	2383	2400	280	100
SVM	2293	2300	380	190
Neural Network	2343	2350	330	140
XGBoost	2393	2450	230	90
DAG Ensemble	2438	2500	180	45

D. Analysis of Experiment Result

The experimental results demonstrate that the proposed DAG-based ensemble model outperforms all baseline models across all evaluated metrics. With an accuracy of 90% and an F1-score of 0.87, the DAG ensemble effectively captures the complexities of the sensor data, making it highly suitable for predictive maintenance tasks.

The Random Forest and XGBoost models also performed well, achieving 85% and 87% accuracy, respectively. However, they could not match the performance of the DAG ensemble, highlighting the advantage of leveraging multiple models and their interactions through a DAG structure.

The ROC-AUC values further emphasize the strength of the DAG ensemble, with a score of 0.94 indicating excellent capability in distinguishing between failure and non-failure cases. The combination of models in the DAG allows for robust handling of various data patterns, enhancing the predictive power of the ensemble.

The confusion matrix analysis reveals that while the DAG ensemble model has a low false positive rate, it exhibits an even lower false negative rate, making it a reliable choice for predictive maintenance in industrial applications. This analysis indicates that integrating diverse machine learning models, mainly through a DAG architecture, improves performance in predictive maintenance scenarios. Future work could explore further optimization of the DAG structure, tuning hyperparameters for individual models, and testing on additional datasets to validate the generalizability of the approach.

V. CONCLUSION

It has been suggested that the proposed DAG-based ensemble learning solution is more flexible and accurate for predictive maintenance in the Industry 4.0 context. DAGs help implement the specialized models since data can be passed dynamically across the models, thus resulting in improved prediction. Future research may consider using dynamic DAGs to represent and adapt the model depending on the new conditions in the industry or using reinforcement learning to select the models. The future work of the current study is to extend the cultivation of the DAGs to be dynamic and self-adaptively changing according to the state of a machine and the use of reinforcement learning for dynamic model ensembling.

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